

Tree growth responses to extreme drought events are not well predicted by climate

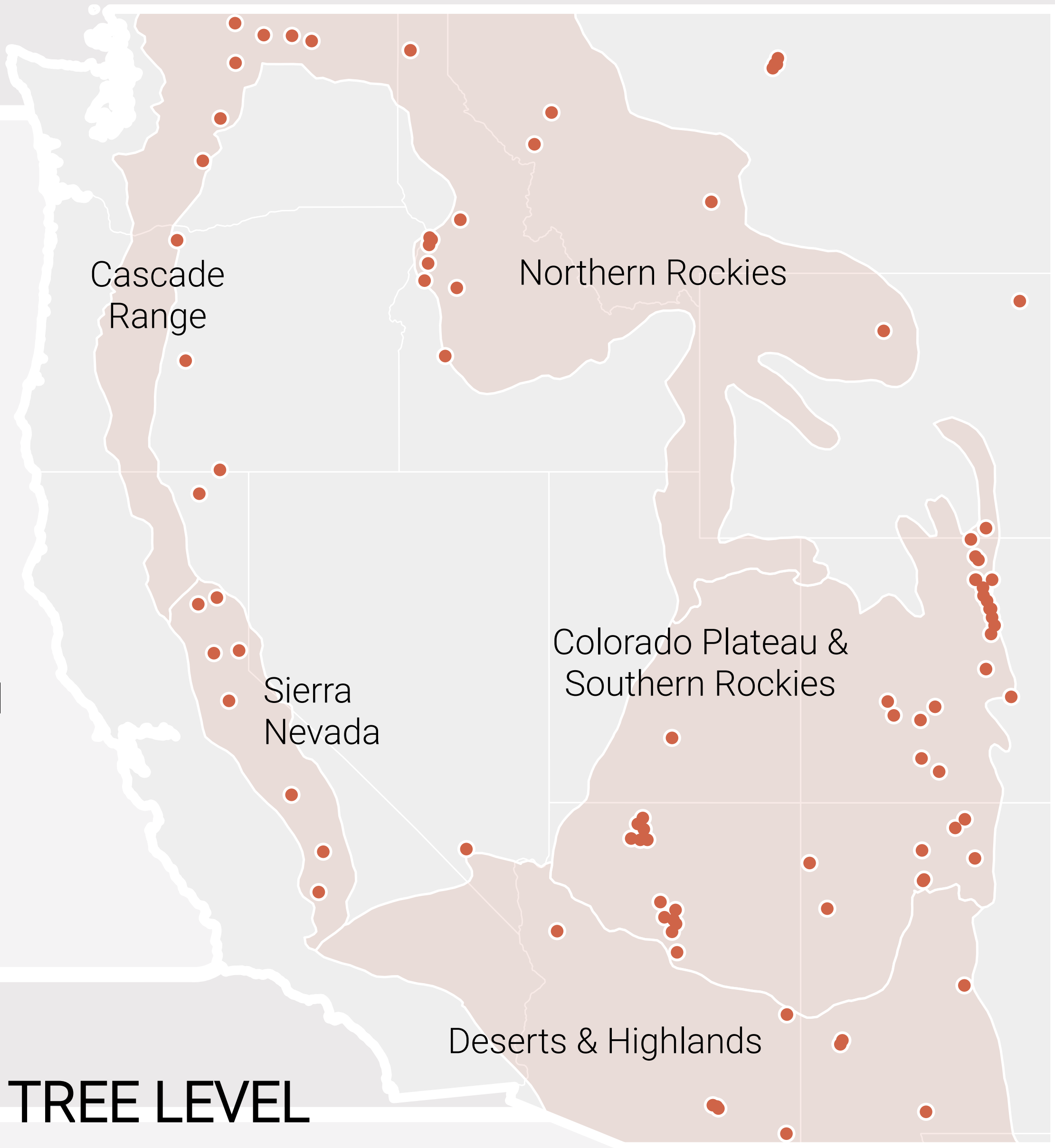
Victor Van der Meersch, Ben Cook, etc.

OVERVIEW

- Climate change is **intensifying droughts** through shifts in temperature, vapor pressure deficit (VPD), and soil moisture
- Combined effects of these drivers on tree growth remain poorly understood at large scales
- A new **Bayesian approach** to jointly model tree growth across Western US, at the tree level
- Our model is able to separate climate effects from other complex dynamics (e.g. allometry)
- Some extreme drought caused **exceptional growth reductions** far exceeding what standard climate variables predict
- Current models likely **underestimate growth declines** during extreme events

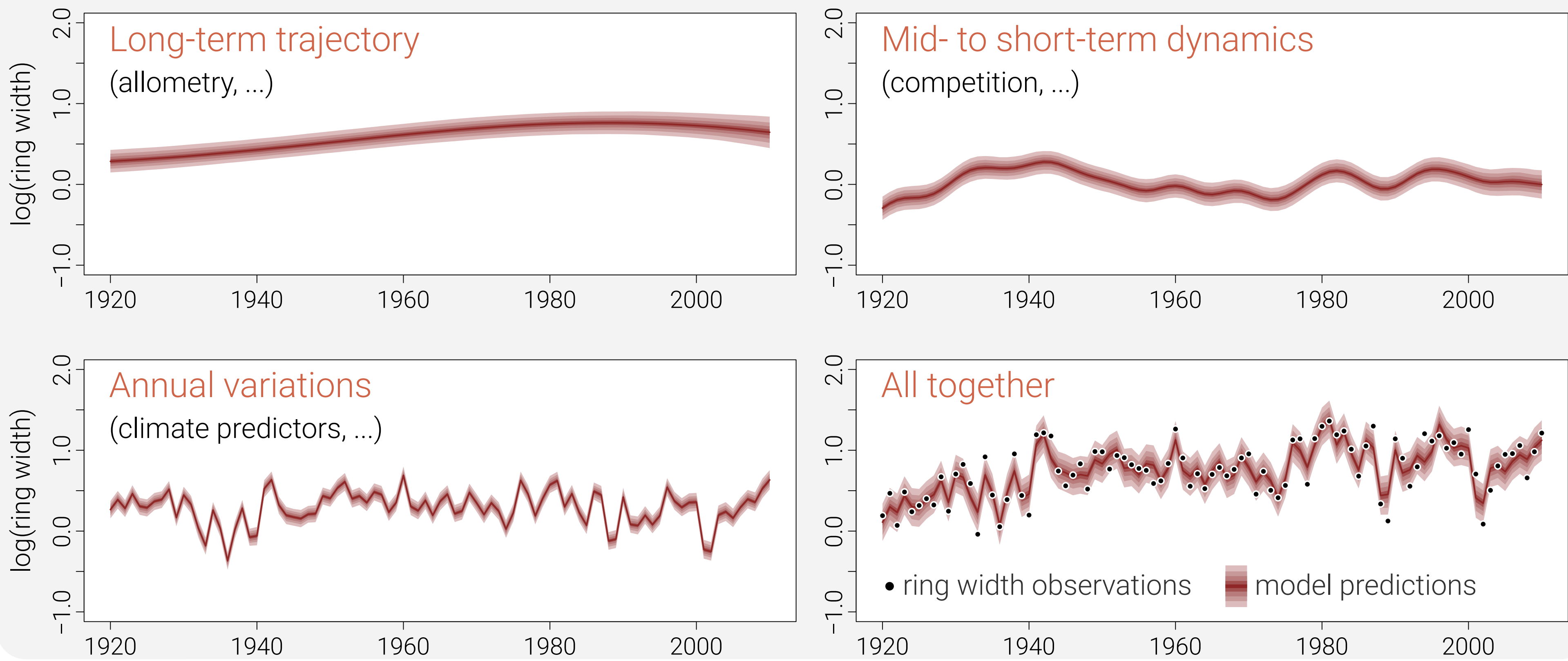
1. BACKGROUND

- Tree-ring width time series are one of the longest and most spatially extensive records of forest response to climate
- However, tree rings are affected by many drivers other than climate: age-size effects, competition, biotic disturbance...
- Standard dendrochronology approaches detrend individual trees and aggregate them to identify some shared climatic signal
- But drought stress operates at the individual level, whose responses integrate non-linear physiological processes
- To capture how extreme events, we need to model individual trees explicitly



2. MODELING GROWTH AT THE TREE LEVEL

Growth is resulting from three major components



3. MAIN CLIMATIC DRIVERS OF GROWTH

Some stuff about the predictors, how much variation they explain...

FIGURE OF CLIMATE PRED IMPACT ACROSS WESTERN US

Highlight some good and bad periods?

4. EXTREME GROWTH REDUCTIONS

Some individual trees have extreme responses

These extreme are often experienced by several trees in a forest

FIGURE : a shocky tree, with predictions from model without shocks?

FIGURE : shocky tree 2

FIGURE : shocky tree 3

FIGURE : shocky tree 3

FIGURE : shocky tree 4

These extreme have major consequences at ecosystem scale

FIGURE OF SHOCKS ACROSS WESTERN US

Blablabla about shocks
% of variation explained

Major droughts

5. IMPLICATIONS FOR FORESTS AND FORECASTS

References...